

Light-source detectives

Teacher guide and worked answers

Distinguish objects that give out their own light from surfaces that reflect light.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsb/inputs/Build SOW 2026-2027.xlsx | BUILD Weekly - Spring | C34

Identify sources of light and surfaces that reflect light.

Direct match to the named outcome. The lesson is a 40-minute exemplar, with optional greater depth; no accreditation completion is claimed.

Prior learning

Everyday objects can be described by what we observe.

Use a labelled diagram or evidence card to explain a choice.

Success evidence

Use evidence to classify a source or reflector.

Explain how a non-glowing object can be seen.

Recognise reflection from both mirrors and ordinary surfaces.

Equipment

Printed object cards; low-power LED torch; white paper; unbreakable mirror; dark fabric; optional switched-on screen.

Preparation

Print the six evidence cards. Check an unbreakable mirror for sharp edges.

If demonstrating reflection, keep the beam on a desk or screen, away from faces.

Practical care

Use a low-power cool LED only. No lasers, hot lamps or direct Sun viewing.

Use an unbreakable mirror and keep reflected light away from eyes. Do not darken the room fully.

Ready to teach alternative

Sort the six supplied cases and trace the supplied mirror and paper diagrams; no practical is needed.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Does bright mean source?
2	Retrieval	2-5	Follow the light
3	I do	5-9	I ask where the light began

Slide	Stage	Minutes	Focus
4	I do	9-12	A mirror redirects light
5	I do	Within stage	Paper reflects too
6	We do	12-16	Sort the evidence cards
7	We do	16-20	The switch-off challenge
8	Hinge	20-23	Which statement works?
9	You do	23-25	Choose your investigation route
10	You do	Within stage	Use the evidence
11	You do	25-35	Make your own science record
12	You do	Within stage	Check and improve
13	Review	35-38	Lundy loop: improve our account
14	Exit	38-40	Three quick checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Does bright mean source?

Invite a prediction without settling it. State that source means gives out its own visible light here. Objects can also reflect while emitting; we classify the stated situation.

2 Retrieval Follow the light

Answers: light; yes; no. If needed trace one source → object → eye path from the previous lesson.

3 I do I ask where the light began

Think aloud: "I do not sort by bright or shiny. I ask whether this object gives out its own light. The lamp does. The plain card returns some arriving light." A source can also reflect; the card's reflected light began elsewhere.

4 I do A mirror redirects light

Think aloud: "I trace the arrows towards the mirror, then away. The mirror changes the direction; it does not switch on its own supply of light." Diagram uses equal incident/reflection angles qualitatively; no angle calculation required.

5 I do Paper reflects too

Shares I do time. Think aloud: "I can see the ordinary paper because light reaches my eyes from it. The arrows leave in several directions. Reflection does not always give a clear image." Arrow counts are not measured amounts.

6 We do Sort the evidence cards

Place two headings on the desk. Partners take reader and sorter roles, swapping after three cards. Use the evidence rather than memory alone. A/C/E source; B/D/F reflector.

Reveal: A, C and E give out their own light in these cases.

7 We do The switch-off challenge

Hold up or describe an unlit phone screen. Let each pair revise one classification. In this stated off case the screen reflects room light, so it can still be visible. Also ask about the mirror in no light: it cannot create a view.

Reveal: The off screen can reflect room light. It no longer supplies its own display light.

8 Hinge Which statement works?

If only mirrors are credited with reflection, return to the paper diagram. If Moon is selected as source, ask pupils to trace the light backwards to the Sun without looking at it.

A. All shiny objects make their own light. - Shiny surfaces such as mirrors reflect arriving light.

B. Only a mirror can reflect light. - Paper and many other surfaces reflect light too.

C. We can see paper when reflected light reaches our eyes. - Paper reflects some light from a source into the eye.

9 You do Choose your investigation route

Give one route only, selected with the pupil. The shared evidence page supports every route. Optional paper-and-mirror observation is supporting evidence; card sorting provides the full independent route. Avoid treating the classes as mutually exclusive for every possible object or condition.

10 You do Use the evidence

Shares the 12-minute independent stage. Optional paper-and-mirror observation is supporting evidence; card sorting provides the full independent route. Avoid treating the classes as mutually exclusive for every possible object or condition.

11 You do Make your own science record

Allow ten minutes of independent response after the two-minute setup. Optional paper-and-mirror observation is supporting evidence; card sorting provides the full independent route. Avoid treating the classes as mutually exclusive for every possible object or condition.

12 You do Check and improve

Shares the independent stage. Ask the pupil to point to the clue supporting their answer; accept a spoken correction or changed choice.

13 Review Lundy loop: improve our account

Give a private or paired option. Adopt one scientifically sound pupil revision in the class explanation and name the change. No public ranking.

Reveal: Ask: does it give out its own light, or reflect light arriving from elsewhere?

14 Exit Three quick checks

Collect brief individual answers. Secure core requires the main idea and a relevant reason. Revisit the relevant worked diagram next lesson if the explanation reverses cause and effect.

Reveal: See the teacher answer guide for the three checks.

Answers and assessment

Retrieval 1-3

Light; light enters the eye; ordinary unlit card does not glow.

We do W1-W2

W1 A, C and E sources; B, D and F reflectors. W2 an off screen can reflect room light although it no longer emits display light.

Hinge

Third option correct. Mirrors do not create the reflected light; reflection also occurs at non-shiny surfaces.

S1-S5

S1 source. S2 reflects. S3 B and F reflectors. S4 source light reaches paper and some reflected light enters the eye. S5 firefly.

T1-T3

T1 A source; B reflector; C source; D reflector; E source; F reflector. T2 mirror returns some arriving light towards the eye. T3 arrows from source to paper and from paper towards eye.

X1-X4

X1 the bright mirror reflects light. X2 ask whether the object gives out its own light in this stated condition. X3 on screen emits display light and can also reflect; off screen can still reflect room light. X4 no: with no incoming light it cannot illuminate the box.

Exit E1-E3

E1 gives out its own light. E2 yes. E3 no, it reflects sunlight.

R1-R2

Responses vary. Credit a science-related revision or relevant next question. A pass or private response is acceptable; preferences have no fixed right answer.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 7; We do 8; hinge 3; You do 12; review 3; exit 2. Zero-minute slides share the stage time.

BUILD follows the user's Year 3 science anchor with age-respectful materials for secondary pupils. Routes are adjustable access choices, not fixed groups or diagnoses.

Every core task can run from the printed evidence. Do not present constructed teaching evidence as observations pupils made. A pupil's spoken reasoning carries the same scientific value as writing.

The word source can be used differently in advanced optics. Here it means a primary visible-light emitter, matching the elementary curriculum distinction. Accept that lit sources can reflect too.

HTML We do: pupils choose a prediction, test it against the user-controlled SVG model or supplied evidence, then explain or revise. The shared paper cases and W1/W2 prompts provide the equivalent classroom discussion without a device. Any animation is a teaching model, not filmed experimental evidence.

Access and responsive teaching

Supported

Choose one route: short choices, word bank and pointing or adult-scribed explanation; keep the same science goal.

Standard

Use the evidence provided to record a decision and explain the science in short sentences.

Stretch

Apply the same core idea to a changed setup, identify a limitation and justify a better decision.

Regulation

Preview the sequence and equipment. Offer seated observation, a quiet paper route, partner rehearsal and a pass. Never require public performance or a darkened room.

Misconceptions to check

Shiny objects always make light.

A shiny surface can reflect light from another source.

Check: Ask what happens when no light reaches the mirror.

Only mirrors reflect light.

Ordinary paper and other visible non-glowing objects reflect light too.

Check: Ask how light from paper reaches the eye.

The Moon is a light source in this classification.

The Moon is seen mainly by reflected sunlight; it does not make that visible light.

Check: Ask where the light began.

Word	Meaning
source	Something that gives out its own light.
reflection	Light returning from a surface.
surface	The outside of an object.

Scientific references

Department for Education: science programmes of study

Checked 8 September 2026. Year 3 light outcomes anchor the BUILD content; ray explanations are carefully modelled as an accessible extension.